XIAMEN UNIVERSITY MALAYSIA

FINAL EXAMINATION

Course Code:	COM104			
Course Name:	Engineering Maths (II)			
Question Paper Setter:	Lohgheswary A/P Nagarethinam			
Academic Session:	2022/04	Question Paper:	A ☒	B ☐
Total No. of Pages:	4	Time Allocated:	2 hours	
Additional Materials:	-			
Apparatus Allowed:	-			

INSTRUCTIONS TO CANDIDATES

1. Answer all questions.
2. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.

DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO.

(Student ID: _____ Full Name: _____)

Question 1 (25 marks, CLO2, PLO1, C3)

(a) Determine the currents I_1, I_2, I_3 in the circuit shown in Figure 1.

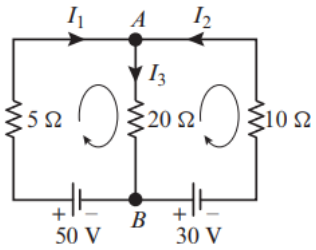


Figure 1

[13 marks]

(b) When chemical compounds are combined under the right conditions, the atoms in their molecules rearrange to form new compounds. For example, when methane burns, the methane CH_4 and stable oxygen O_2 react to form carbon dioxide CO_2 and water H_2O. This is indicated by the chemical equation

$$CH_4 + O_2 \rightarrow CO_2 + H_2O$$

Balance the equation.

[12 marks]

Question 2 (25 marks, CLO1, PLO1, C2)

(a) Evaluate $\det(A)$ where $A = \begin{bmatrix} 0 & 1 & 5 \\ 3 & -6 & 9 \\ 2 & 6 & 1 \end{bmatrix}$.

[9 marks]

(b) Evaluate $\det(A)$ where $A = \begin{bmatrix} 3 & 5 & -2 & 6 \\ 1 & 2 & -1 & 1 \\ 2 & 4 & 1 & 5 \\ 3 & 7 & 5 & 3 \end{bmatrix}$.

[8 marks]

(c) Find a basis for the row space of $A = \begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 2 & -5 & -3 & -2 & 6 \\ 0 & 5 & 15 & 10 & 0 \\ 2 & 6 & 18 & 8 & 6 \end{bmatrix}$ consisting entirely of row vectors from A.

[8 marks]

Question 3 (25 marks, CLO2, PLO1, C3)

(a) Assume that the vector space R^3 has the Euclidean inner product. Apply the Gram-Schmidt process to transform the basis vectors $\vec{u}_1 = (1,1,1), \vec{u}_2 = (0,1,1), \vec{u}_3 = (0,0,1)$ into

(i) an orthogonal basis $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ [6 marks]

(ii) an orthonormal basis $\{\vec{q}_1, \vec{q}_2, \vec{q}_3\}$ [7 marks]

(b) P_2 have the evaluation inner product at the points $x_0 = -2$, $x_1 = 0$ and $x_2 = 2$. Compute

(i) $\langle p, q \rangle$ [4 marks]

(ii) $\|p\|$ [2 marks]

For the polynomials $p = p(x) = x^2$ and $q = q(x) = 1 + x$.

(c) R^3 have the Euclidean inner product and W be the subspace spanned by the orthonormal vectors $v_1 = (0,1,0)$ and $v_2 = (-\frac{4}{5}, 0, \frac{3}{5})$. Find

(i) orthogonal projection of $u = (1,1,1)$ on W [3 marks]

(ii) the component of u orthogonal to W [3 marks]

Question 4 (25 marks, CLO1, PLO1, C2)

Find

(a) eigenvalues for $A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{bmatrix}$. [8 marks]

(b) an orthogonal matrix P that diagonalizes $A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{bmatrix}$. [17 marks]

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